Submission 2: Vitamin D and Gene Expression

Chunlei Li

Research question

I continued my Submission 1 analysis of CYP24A1 with baseline vitamin D (continuous), pregnancy outcome (categorical), and vitamin D group (categorical, derived from vitamin D using 30 ng/mL). I added UTS2 and RUNDC3A, the two genes with the highest variance among the 500 genes.

Data preparation

```
# Read the original files
covar <- read.csv("GSE85307_covar-1.csv")
expr <- read.csv("GSE85307_geneExpression.csv", row.names = 1)
geneNames <- read.csv("GSE85307_geneNames.csv")
```

```
# Remove the text before ": " in the covariates I use
covar <- covar %>%
  separate(col = preg_preeclampsia, into = c(NA, "preg_preeclampsia"),
           sep = ": ") %>%
  separate(col = mat_vitD, into = c(NA, "mat_vitD"), sep = ": ")
```

```
# Convert vitamin D to a number and pregnancy outcome to a factor
covar$mat_vitD <- as.numeric(covar$mat_vitD)
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c("Normal", "Preeclampsia"))
```

```
# Derive the vitamin D group (cutoff 30 ng/mL)
covar$vitD_group <- factor(ifelse(covar$mat_vitD < 30,
                                  "<30 ng/mL", ">=30 ng/mL"))

# Order expression columns by geoID and check the match
expr <- expr[, covar$geoID]
table(colnames(expr) == covar$geoID)
```

```
##
## TRUE
##  157
```

```
# Replace probe IDs with gene names and check for missing names
row.names(geneNames) <- geneNames$PROBEID
row.names(expr) <- geneNames[row.names(expr), "GENE"]
sum(is.na(row.names(expr)))
```

```
## [1] 0
```

```
# One row per participant: covariates plus all genes
analysisData <- cbind(covar, t(expr))
dim(analysisData)
```

```
## [1] 157 511
```

```
# Subgroup sizes
table(analysisData$preg_preeclampsia, analysisData$vitD_group)
```

```
##
##                <30 ng/mL >=30 ng/mL
##   Normal              78         32
##   Preeclampsia        41          6
```

Select two additional genes

```
# Variance of each gene, then the two highest
geneVar <- apply(expr, MARGIN = 1, FUN = var)
names(geneVar)[order(geneVar, decreasing = TRUE)[1:2]]
```

```
## [1] "UTS2"    "RUNDC3A"
```

Plotting function

The function takes a data frame and a list of one or more gene names. For each gene it makes a histogram, a scatterplot against vitamin D, and a boxplot by pregnancy outcome and vitamin D group.

```
makeGenePlots <- function(data, genes) {
  # Somewhere to store the plots
  plots <- list()

  # Loop through all genes
  for (gene in genes) {
    # Copy the current gene into one column
    data$Expression <- data[, gene]

    # Histogram
    histPlot <- ggplot(data, aes(x = Expression)) +
      geom_histogram(bins = 24, fill = "#337C8E", color = "white") +
      labs(title = gene, x = paste(gene, "expression"),
           y = "Number of participants") +
      theme_classic()

    # Scatterplot
    scatterPlot <- ggplot(data, aes(x = mat_vitD, y = Expression)) +
      geom_point(color = "#337C8E") +
```

```
      geom_smooth(method = "lm", color = "#B65E27") +
      labs(title = gene, x = "Baseline vitamin D (ng/mL)",
           y = paste(gene, "expression")) +
      theme_classic()

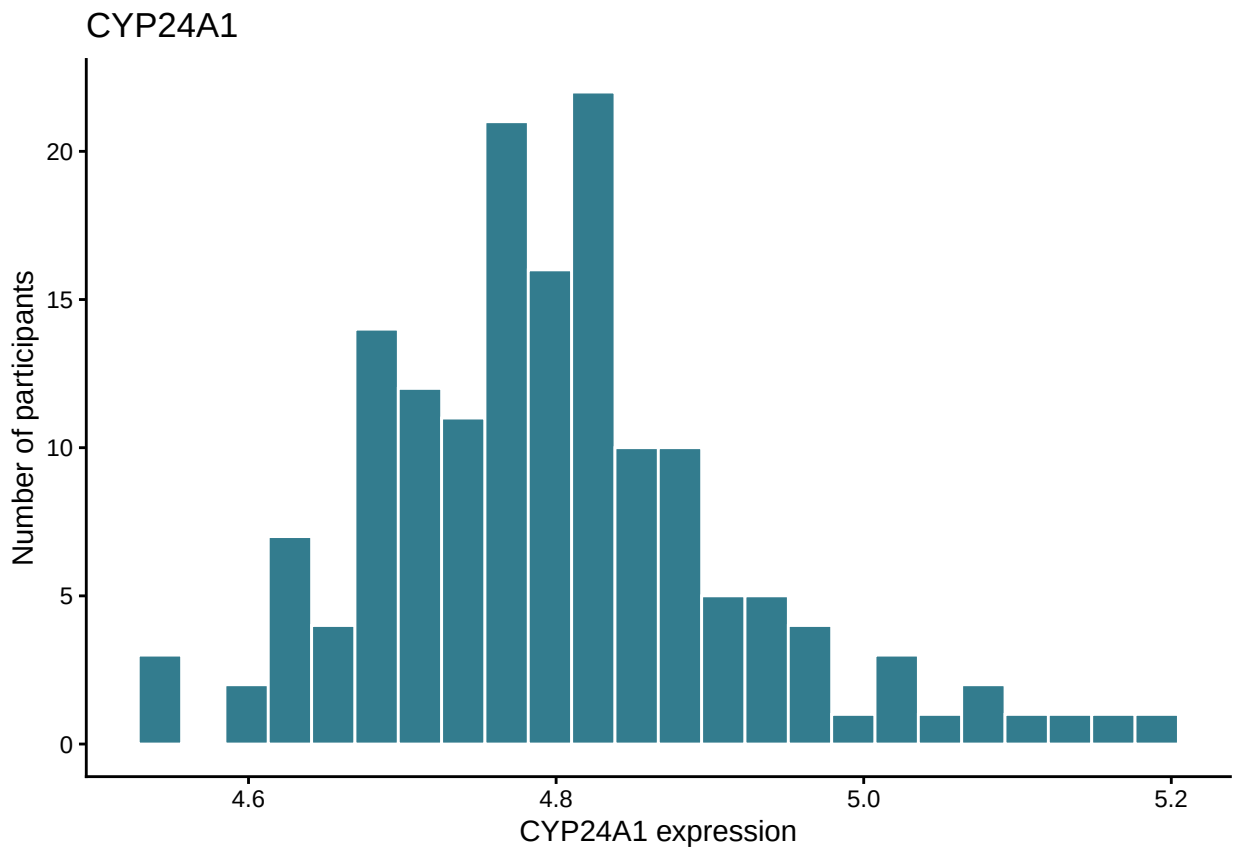
    # Boxplot
    boxPlot <- ggplot(data, aes(x = preg_preeclampsia, y = Expression,
                                fill = vitD_group)) +
      geom_boxplot() +
      scale_fill_manual(values = c("#337C8E", "#D5A13B")) +
      labs(title = gene, x = "Pregnancy outcome",
           y = paste(gene, "expression"), fill = "Baseline vitamin D") +
      theme_classic() +
      theme(legend.position = "top")

    # Save the three plots for this gene
    plots[[gene]] <- list(hist = histPlot, scatter = scatterPlot,
                          box = boxPlot)
  }
  plots
}


plots <- makeGenePlots(data = analysisData,
                       genes = c("CYP24A1", "UTS2", "RUNDC3A"))
```

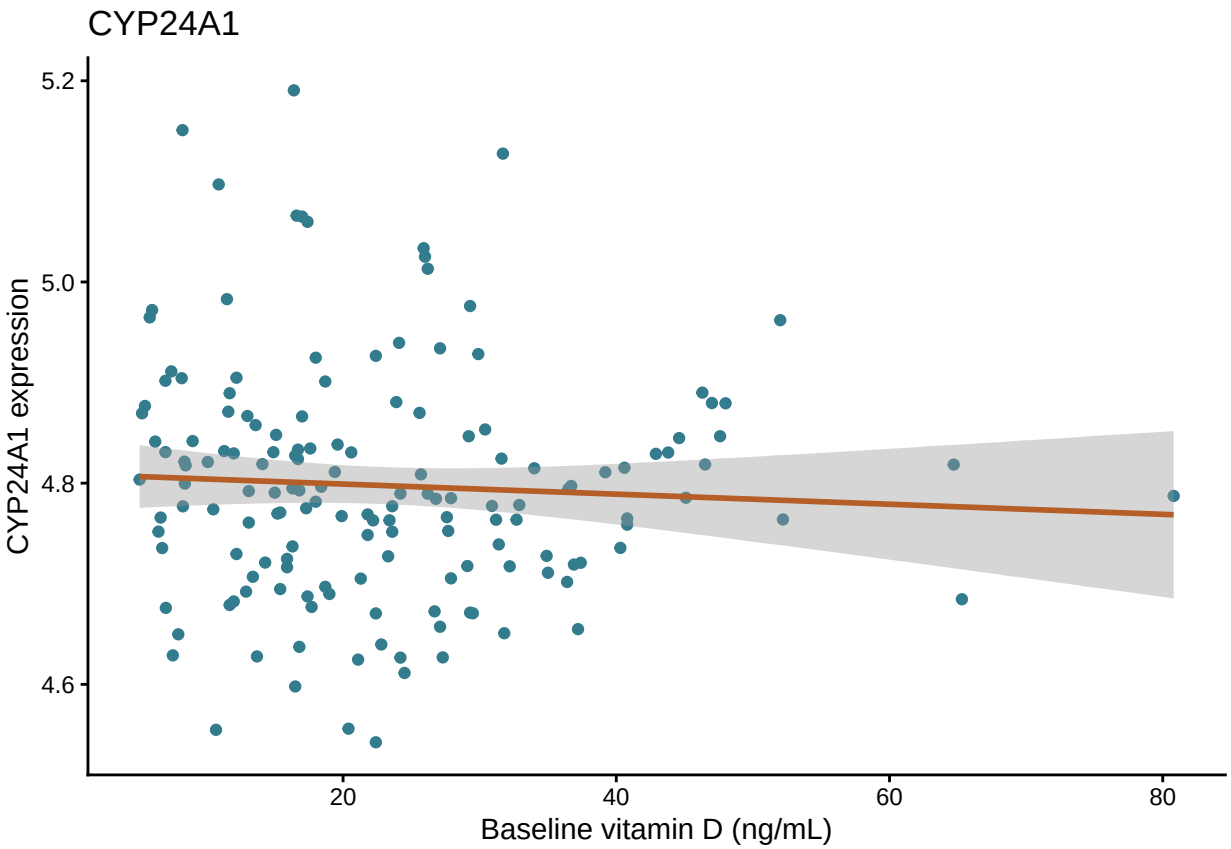
CYP24A1

```
plots$CYP24A1$hist
```



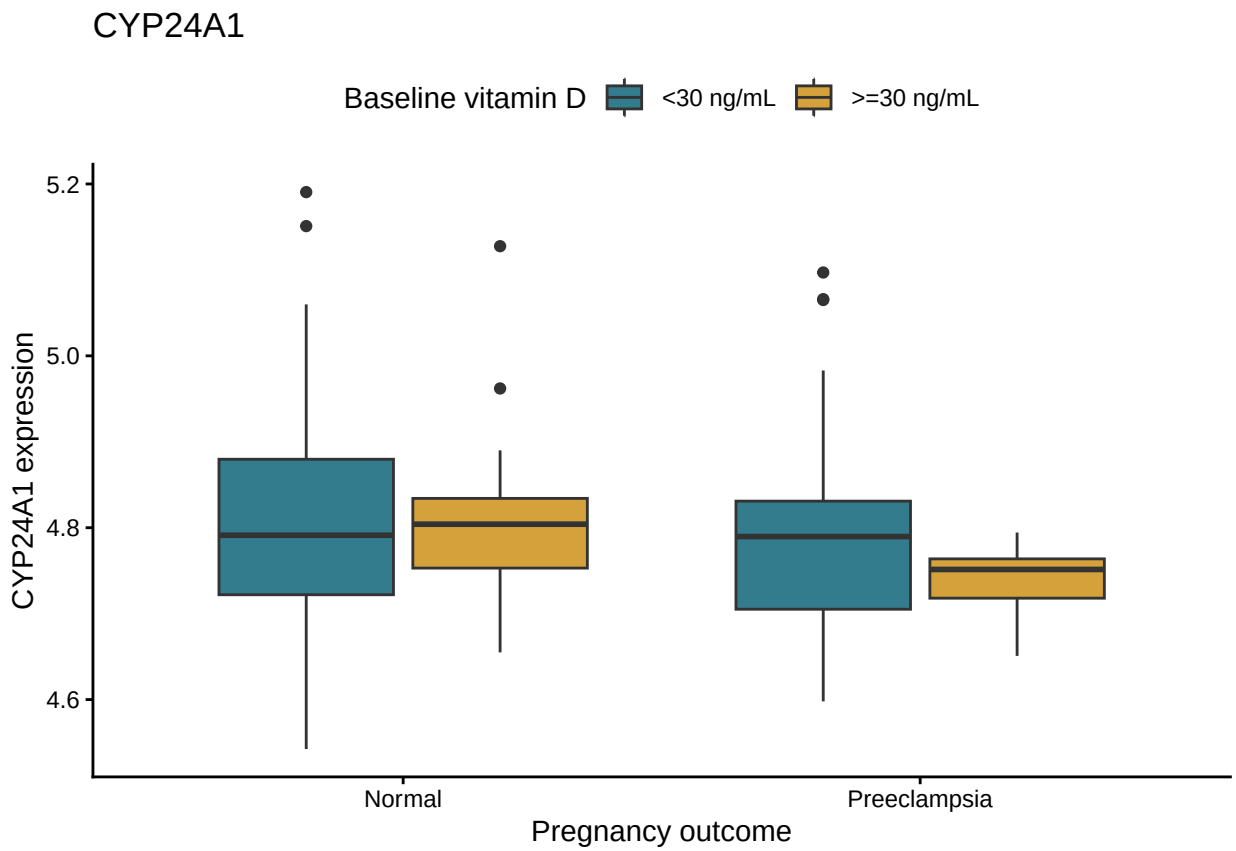
CYP24A1 expression showed one main peak near 4.8 with a small upper tail.

```
plots$CYP24A1$scatter
```



The fitted line was nearly flat, so vitamin D showed no clear linear pattern with CYP24A1 expression.

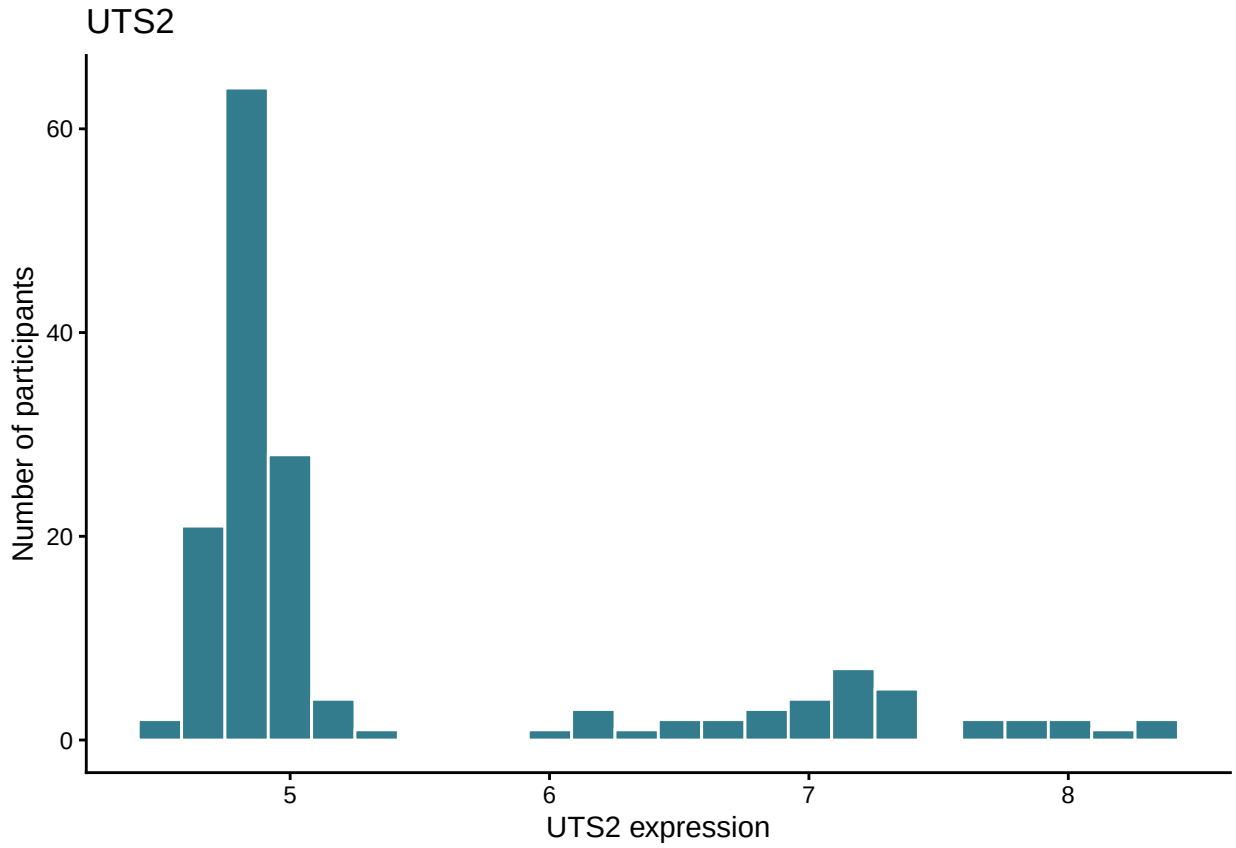
```
plots$CYP24A1$box
```



The distributions overlapped. The preeclampsia group with vitamin D of at least 30 ng/mL had only six participants.

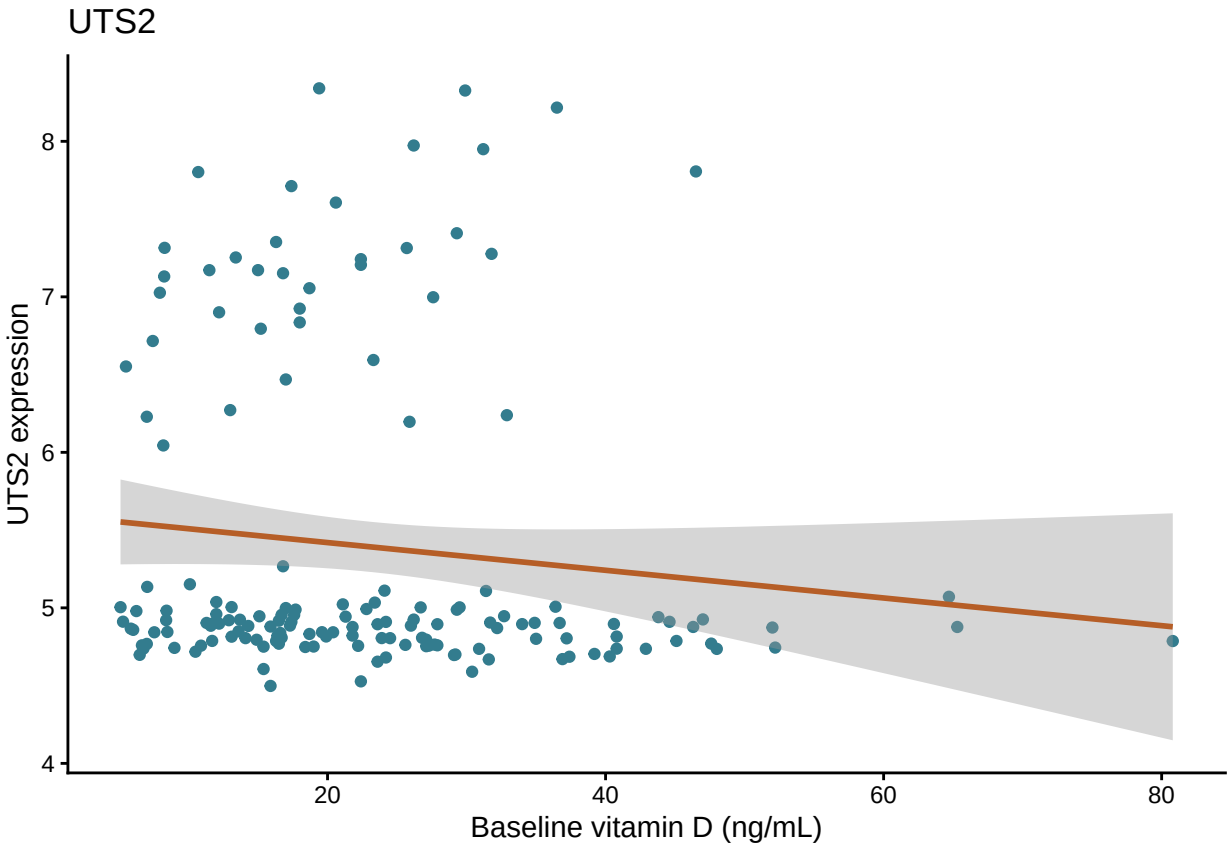
UTS2

```
plots$UTS2$hist
```



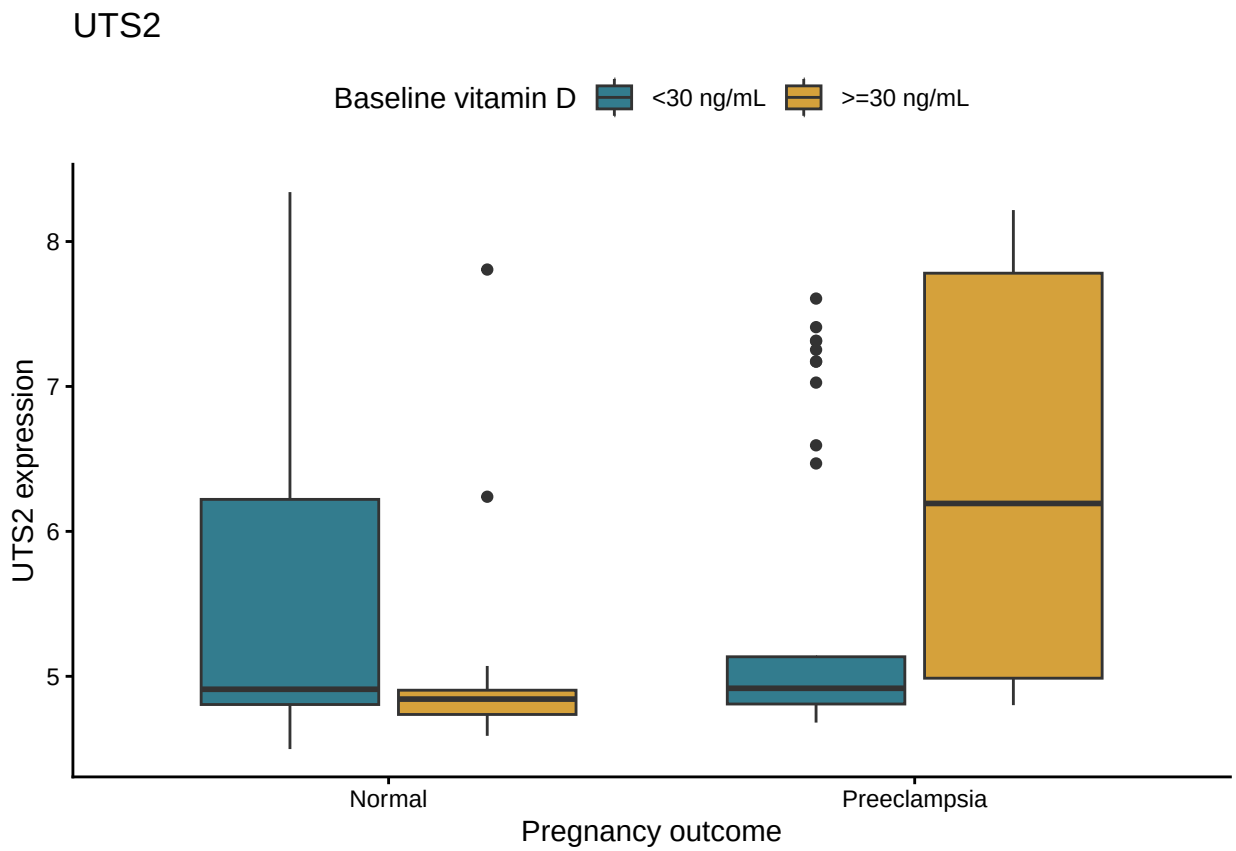
UTS2 expression had two groups: most values were near 5, and a smaller group was between about 6 and 8.5.

```
plots$UTS2$scatter
```



Both UTS2 groups appeared across the range of vitamin D, so the single fitted line did not describe either group well.

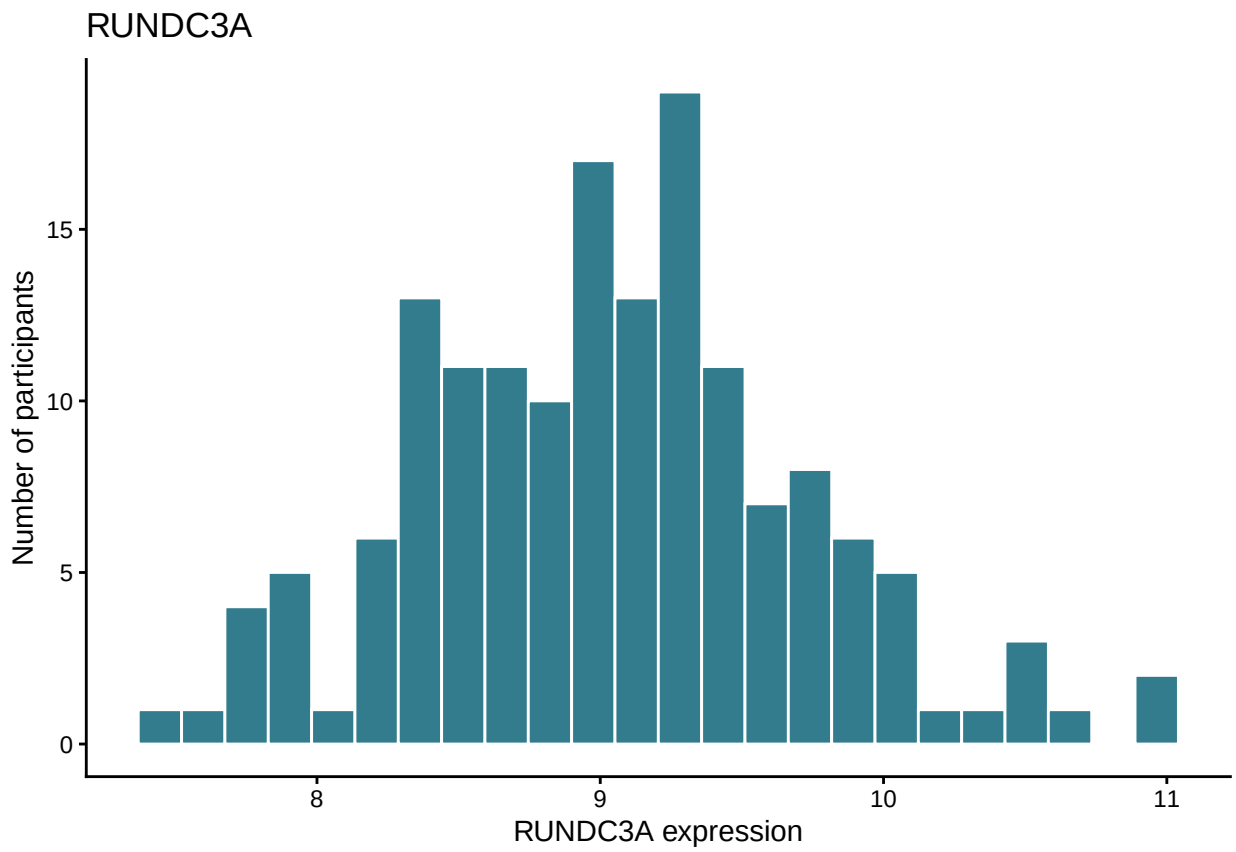
```
plots$UTS2$box
```



The median was higher in the six preeclampsia participants with vitamin D of at least 30 ng/mL, but this subgroup is very small.

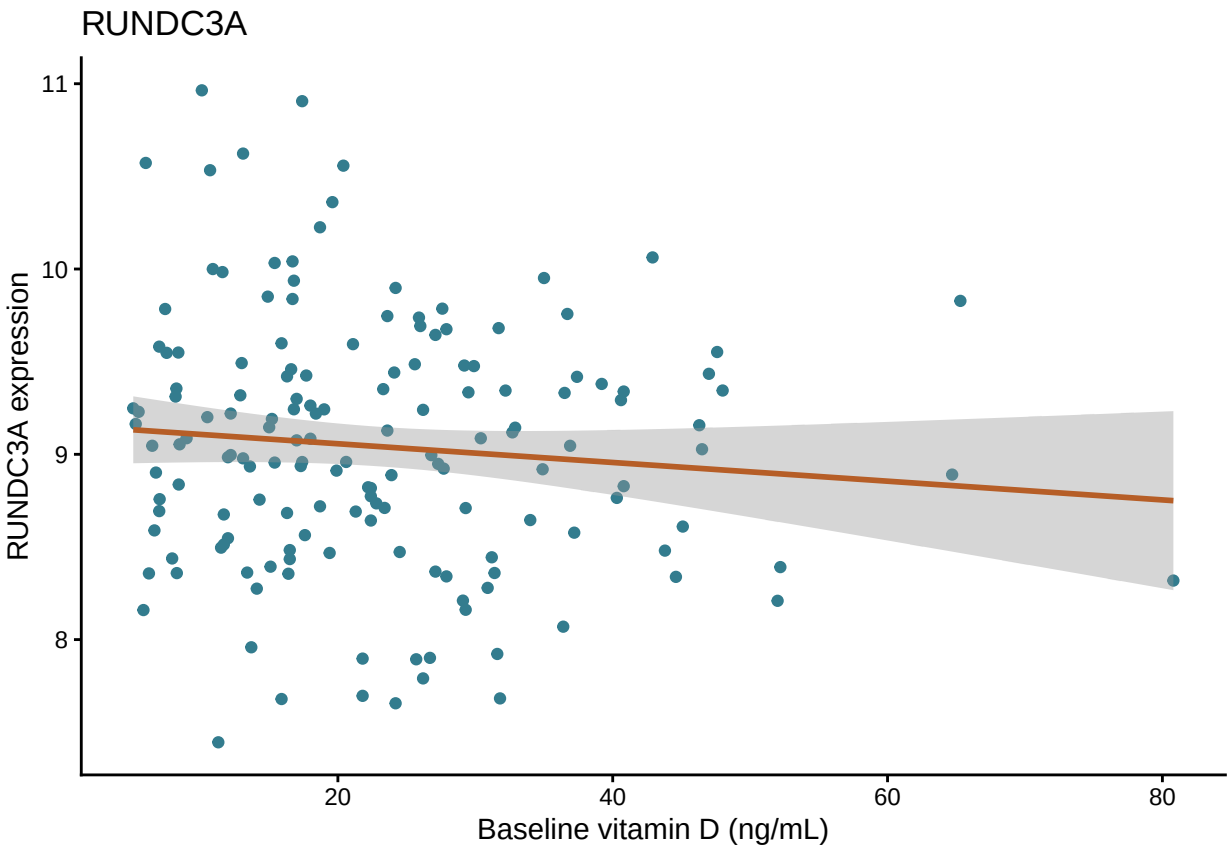
RUNDC3A

```
plots$RUNDC3A$hist
```



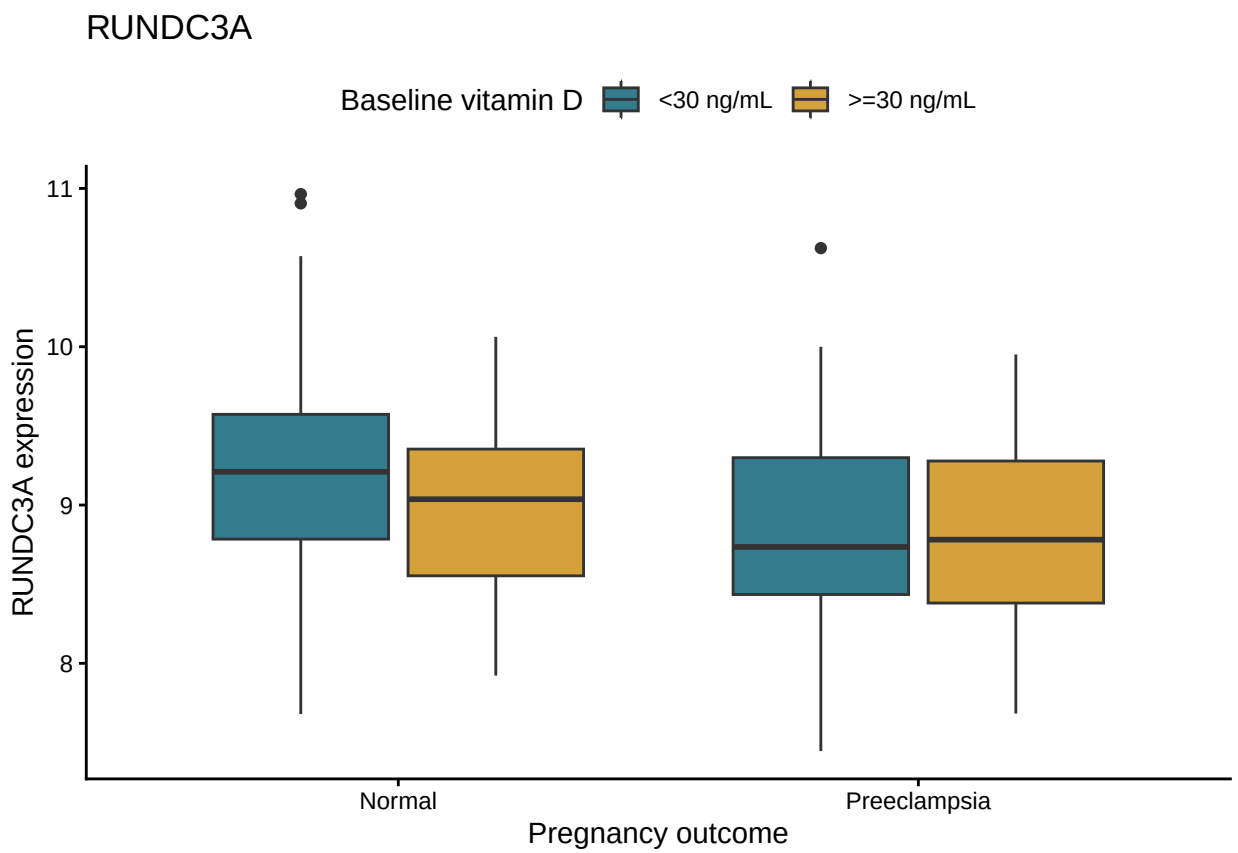
RUNDC3A expression had a broader distribution centered near 9.

```
plots$RUNDC3A$scatter
```



The fitted line sloped slightly downward, with wide scatter.

```
plots$RUNDC3A$box
```



Median RUNDC3A expression was lower in the preeclampsia group within both vitamin D groups, although the distributions overlapped.